

YOLO Model Testing Report

Model:	yolo12x
Model Size:	113.8 MB
Parameters:	59.2 M
FLOPs (640x640):	200.33 GFLOPs
Run Name:	yolo12x_testing_20251124_005556
W&B Run Name:	yolo12x_bdd100k_yolo_test_20251124_005556
Timestamp:	2025-11-24 01:16:30
Dataset:	bdd100k_yolo - test split
Images Processed:	20000
Total Execution Time	1205.80s
IoU Threshold:	0.5

Inference Performance

Metric	Value
Average Inference Time	57.07 ms per image
FPS (Frames Per Second)	17.52

Note: Inference time includes preprocessing, model inference, and postprocessing.

Overall Accuracy Metrics

Metric	Value
Precision	0.3046
Recall	0.6625
F1-Score	0.4173
mAP@0.5	0.1050

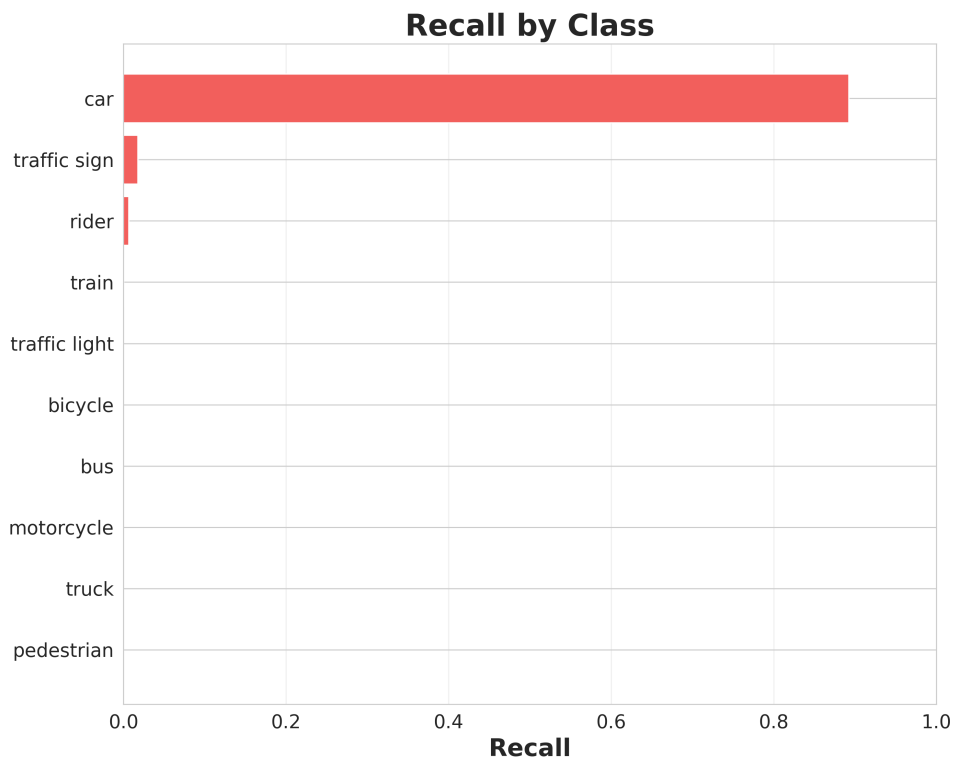
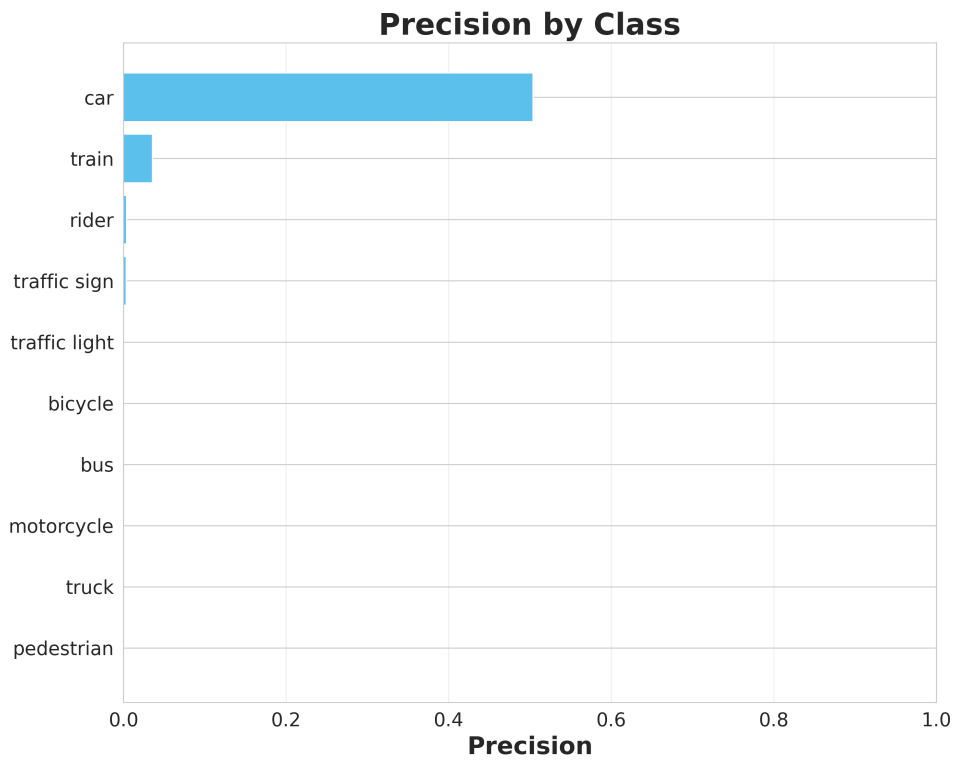
mAP@0.5:0.95	0.0635
True Positives	103608
False Positives	236533
False Negatives	52779

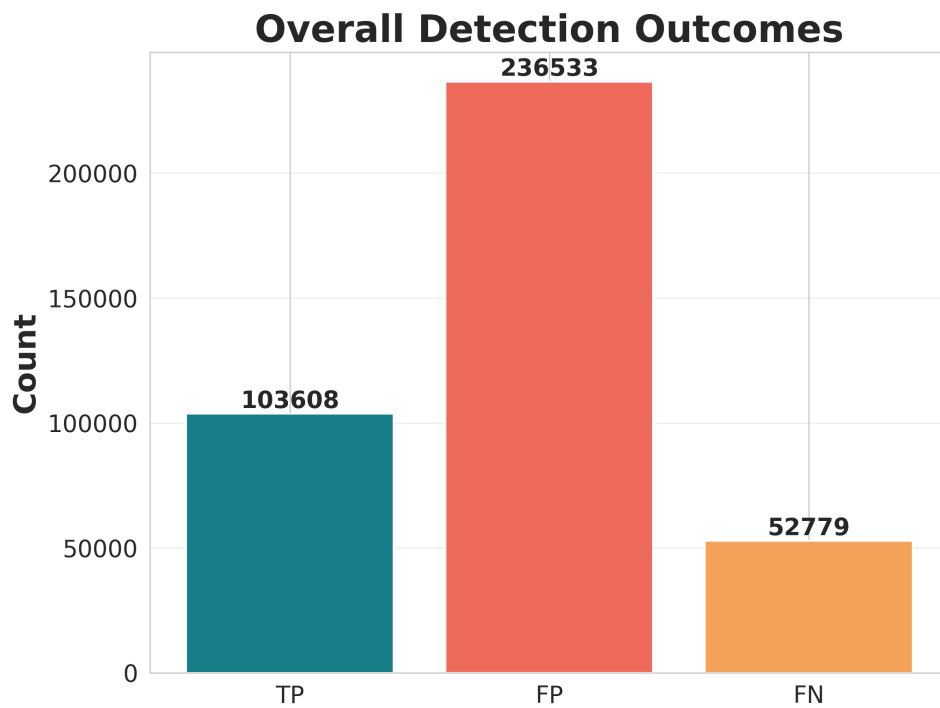
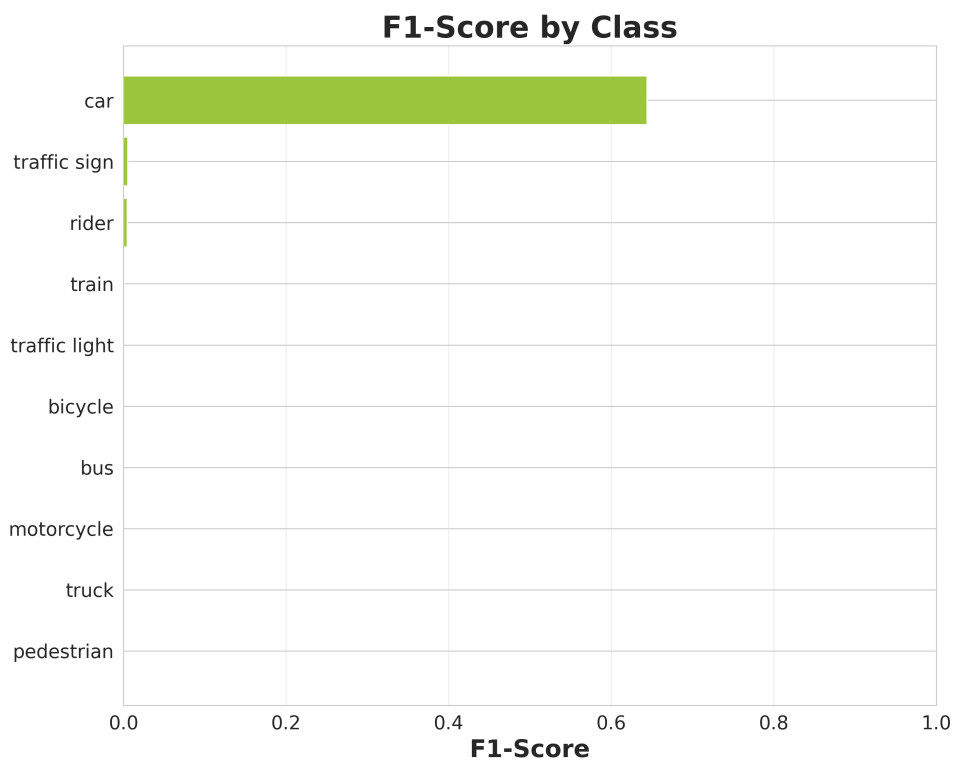
Confusion Matrix

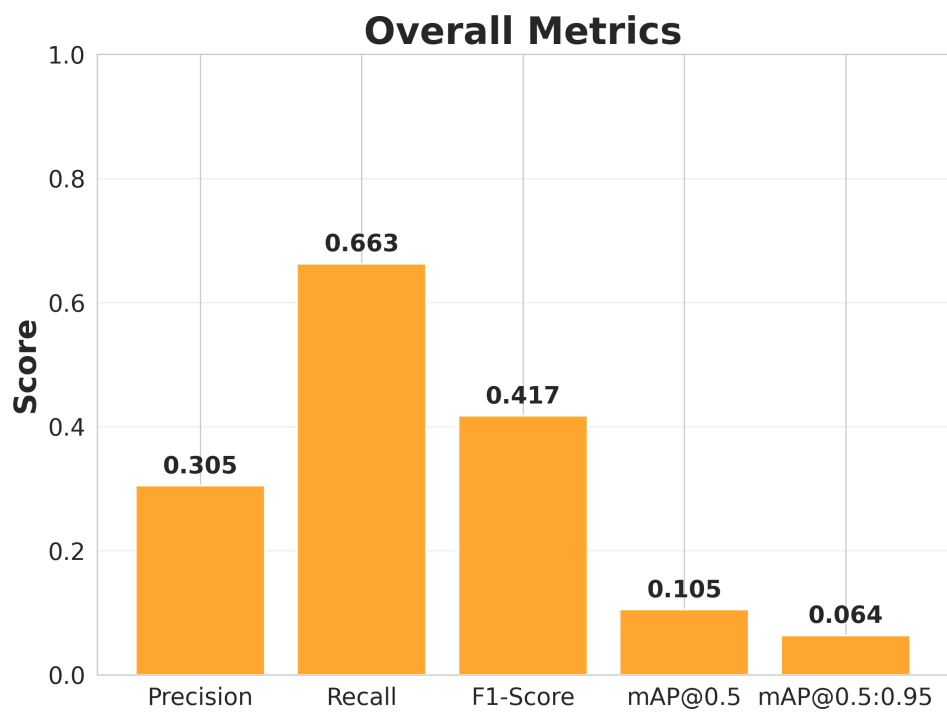
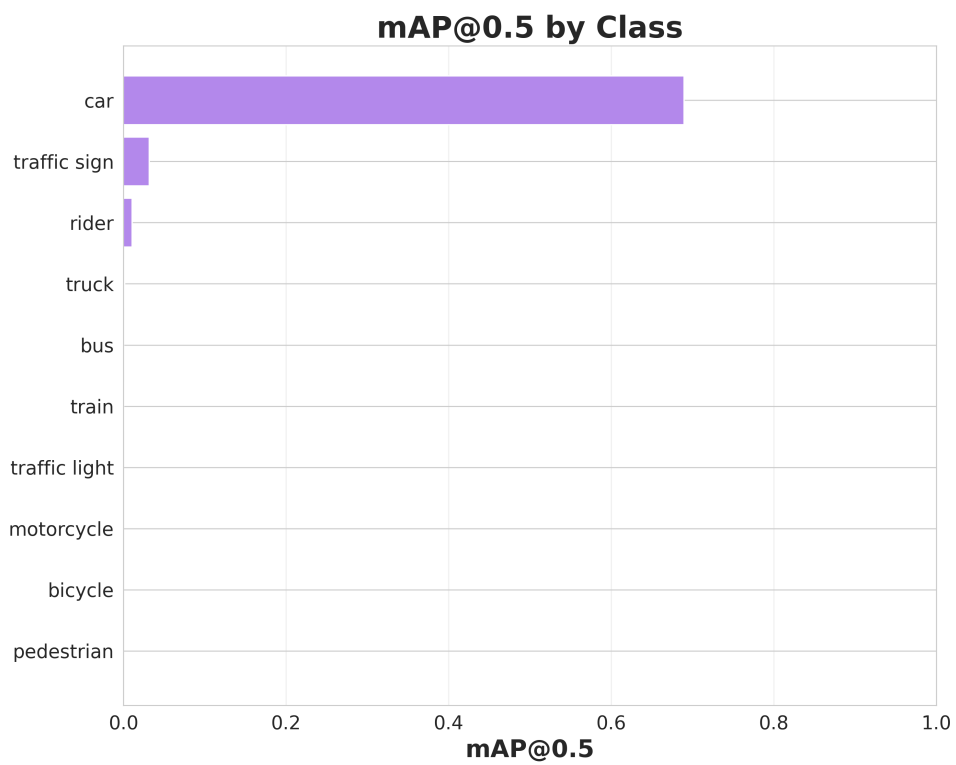
Confusion Matrix (yolo12x validation)											
True Class	pedestrian	667	62	1					4	14	
	rider	5	2								
	car	10	103368	738	166				20	56	
	truck	8	10								
	bus			1							
	train		364	329	1428	1				3	
	motorcycle		5	5	9	8					
	bicycle	1	3707	4233	357	2			1	9	
	traffic light			1							
	traffic sign		1	1		3			9316	234	
		pedestrian	rider	car	truck	bus	train	motorcycle	bicycle	traffic light	traffic sign
Predicted Class											

Correct Predictions (Diagonal Sum): 103608
Total Matched Predictions: 373073

Performance Visualizations







Per-Class Performance

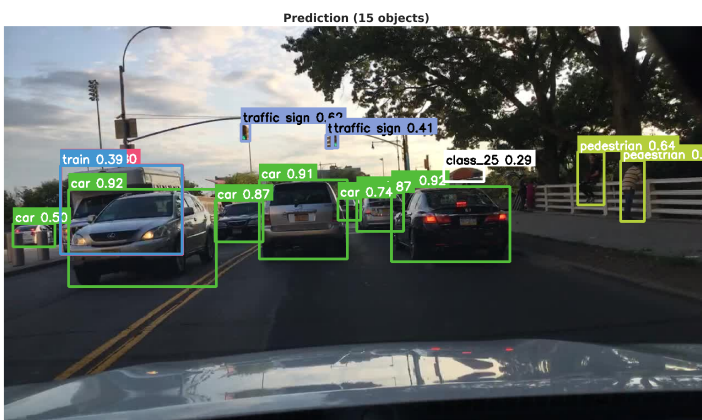
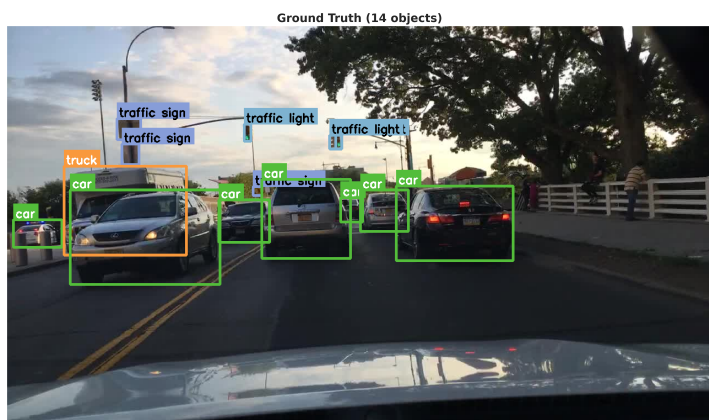
Class	TP	FP	FN	Precision	Recall	F1-Score	mAP@0.5
pedestrian	0	0	12405	0.0000	0.0000	0.0000	0.0000
rider	5	1288	751	0.0039	0.0066	0.0049	0.0107
car	103368	101725	12465	0.5040	0.8924	0.6442	0.6899
truck	0	8701	468	0.0000	0.0000	0.0000	0.0016
bus	0	3217	22	0.0000	0.0000	0.0000	0.0005
train	1	27	2640	0.0357	0.0004	0.0007	0.0005
motorcycle	0	0	183	0.0000	0.0000	0.0000	0.0000
bicycle	0	0	11057	0.0000	0.0000	0.0000	0.0000
traffic light	0	52817	10	0.0000	0.0000	0.0000	0.0000
traffic sign	234	68758	12778	0.0034	0.0180	0.0057	0.0321

Sample Predictions: Ground Truth vs Model

Sample #1: f1d169ec-aebab59c.jpg

Ground Truth Objects: 14 | Predicted Objects: 15

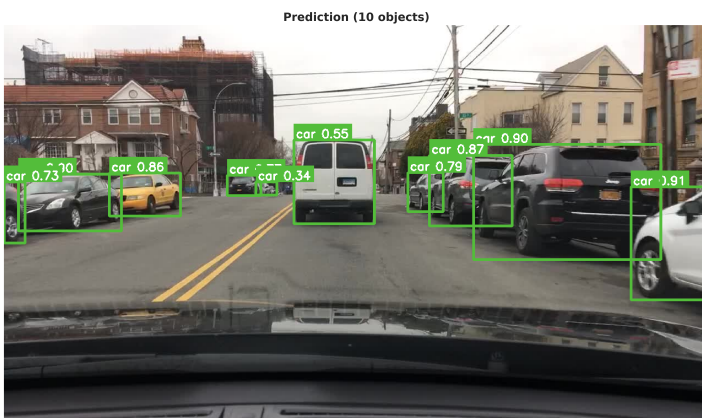
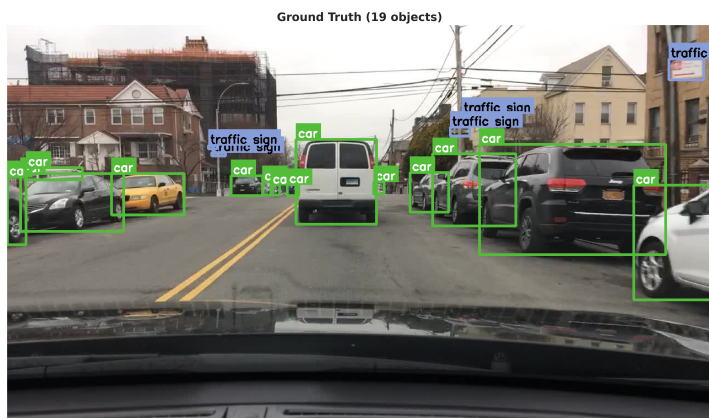
Weather: overcast | Scene: highway | Time of Day: dawn/dusk



Sample #2: dc6936a6-23966dbc.jpg

Ground Truth Objects: 19 | Predicted Objects: 10

Weather: overcast | Scene: residential | Time of Day: daytime



Sample #3: ed024ff2-5243e710.jpg

Ground Truth Objects: 21 | Predicted Objects: 13

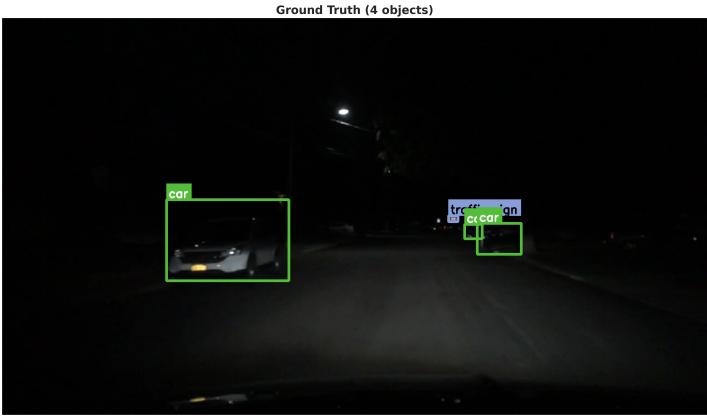
Weather: clear | Scene: city street | Time of Day: daytime



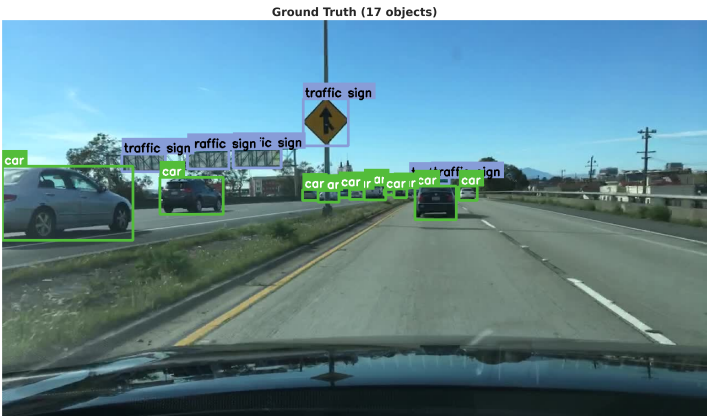
Sample #4: d0f5f208-1b00e4c9.jpg

Ground Truth Objects: 4 | Predicted Objects: 2

Weather: clear | Scene: undefined | Time of Day: night



Sample #5: cb97debb-12f48570.jpg
Ground Truth Objects: 17 | Predicted Objects: 12
Weather: partly cloudy | Scene: highway | Time of Day: daytime



Sample #6: ec2734b1-d5046636.jpg
Ground Truth Objects: 26 | Predicted Objects: 10
Weather: clear | Scene: city street | Time of Day: night

